Ex. No: 1

Title: Vector and Matrix Operations for Deep Learning

Packages: NumPy, Matplotlib

AIM:

To perform vector and matrix operations such as addition, dot product, norm, multiplication, transpose, determinant and inverse using NumPy, and to plot vectors using Matplotlib.

ALGORITHM:

1. Start the program and import NumPy and Matplotlib.
2. Create two vectors a and b.
3. Find a + b, the dot product and the norm of a.
4. Create two matrices A and B.
5. Find A + B, A x B, transpose, determinant and inverse of A.
6. Plot the vectors a and b using Matplotlib.
7. Display the output and stop.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt

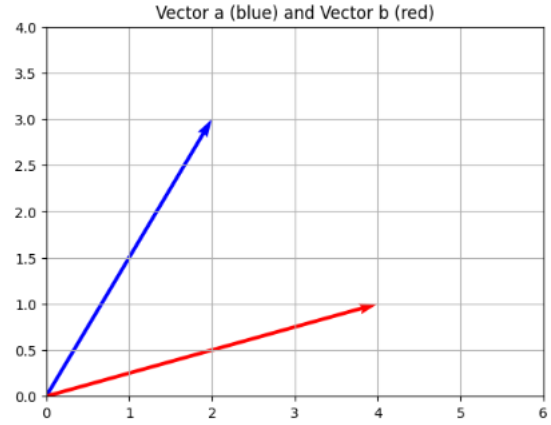
a = np.array([2, 3])
b = np.array([4, 1])
print("a + b =", a + b)
print("Dot product =", np.dot(a, b))
print("Norm of a =", np.linalg.norm(a))

A = np.array([[1, 2], [3, 4]])
B = np.array([[2, 0], [1, 2]])
print("A + B =\n", A + B)
print("A x B =\n", A @ B)
print("Transpose of A =\n", A.T)
print("Determinant of A =", round(np.linalg.det(A), 2))
print("Inverse of A =\n", np.linalg.inv(A))

plt.quiver(0, 0, a[0], a[1], angles='xy', scale_units='xy', scale=1, color='blue')
plt.quiver(0, 0, b[0], b[1], angles='xy', scale_units='xy', scale=1, color='red')
plt.xlim(0, 6)
plt.ylim(0, 4)
plt.grid(True)
plt.title("Vector a (blue) and Vector b (red)")
plt.show()
```

OUTPUT:

```
a + b = [6 4]
Dot product = 11
Norm of a = 3.605551275463989
A + B =
 [[3 2]
 [4 6]]
A x B =
 [[ 4  4]
 [10  8]]
Transpose of A =
 [[1 3]
 [2 4]]
Determinant of A = -2.0
Inverse of A =
 [[-2.   1. ]
 [ 1.5 -0.5]]
```



RESULT:

Vector and matrix operations were performed successfully using NumPy and the vectors were plotted using Matplotlib.